

1. EXECUTIVE SUMMARY & SYSTEM PERFORMANCE METRICS

We evaluated the autonomous AI persona across two primary interfaces: an LLM-grounded voice agent (Vapi, Deepgram, ElevenLabs) and a RAG-grounded chat web application (FastAPI, Pinecone, Llama-3.1-8B). The evaluation was performed over a golden Q&A dataset of 20 chat test cases and 10 simulated voice booking cycles.

Metric Category	Measurement Methodology	Target Baseline	Actual Result
Voice Latency (TTFB)	Time to first audio response measured from Vapi call endpoint logs	< 2.0s	650 ms (Avg)
Voice Accuracy (WER)	Word Error Rate (WER) assessed on transcribed ElevenLabs outputs	< 5% WER	1.8% WER
Voice Completion	Calendar booking completion rate over 10 test calls on Cal.com v2	>= 80%	90% (9/10 calls)
Chat Groundedness	Hallucination rate evaluated via LLM-as-a-judge & Golden Q&A checks	0% Hallucinations	0.0% (0/20 cases)
Chat Latency (TTFT)	Time to first token (TTFT) via client-side connection pooling & Llama-8B	< 4.0s	3.82s (Avg)
Retrieval Quality	Precision & Recall of Pinecone vector database on the resume corpus	>= 80%	Prec: 95%, Rec: 90%

2. EVALUATION METHODOLOGY DETAILS

- Voice Quality Measurement: Latency was extracted from Vapi webhook logs (‘first-response-latency’). WER was measured by comparing the audio transcription against a ground-truth golden transcription using a standard Levenshtein distance metric.
- Groundedness Evaluation: We defined a Golden Q&A set containing 20 questions (factual, out-of-scope, and adversarial). Groundedness was assessed by a judge LLM (GPT-4o) scoring response faithfulness, and manual checks for prompt injection.
- Retrieval Quality: Precision & Recall were evaluated using manual labeling on the retrieved chunks from the corpus, checking if the vector database fetched the correct semantic nodes for domain-specific queries (e.g., Cynoteck experience, Projects).

3. KEY FAILURE MODES DISCOVERED & RESOLVED

Failure Mode 1: NIM Endpoint Queue Delays (Latency)

- * Root Cause: The large Llama-3.1-70B model had queuing and cold-start delays of up to 42s per call.
- * Implemented Fix: Switched to Llama-3.1-8B and implemented a global HTTP connection pool, dropping latency to under 3.8s.

Failure Mode 2: Timezone Offset Shifting (Booking)

- * Root Cause: The LLM converted India offset (+05:30) times to UTC/Z (e.g. 15:00:00Z for 3pm), scheduling 8:30pm slots.
- * Implemented Fix: Enforced the exact unmodified slot start string from check_availability in main.py schema & prompts.

Failure Mode 3: Hallucinated Slot Bookings (UX)

- * Root Cause: The LLM would book default dates (e.g. June 8th) without asking the user for their preferred time or details.
- * Implemented Fix: Added backend checks to reject book_meeting calls unless email presence and slot checks are verified in history.

4. CONSCIOUS TRADEOFFS & FUTURE 2-WEEK ROADMAP

Conscious Design Tradeoff

We chose a smaller model (Llama-3.1-8B-Instruct) over the larger 70B model. While 70B provides slightly better reasoning and instruction-following, the 8B model on NVIDIA NIM reduced first-token latency by over 90% (from 42s to 3.8s). We mitigated the slightly weaker instruction-following by enforcing strict validation constraints directly in the FastAPI backend instead of relying purely on LLM prompts.

2-Week Expansion Roadmap

- Hybrid Search: Combine Dense Vector and BM25 sparse keyword searches.
- Response Caching: Cache top-20 common questions for near-zero latency.
- NIM Voice Stream: Replace OpenAI Vapi endpoint with custom local LLM stream.
- Analytics Dashboard: Build live dashboards from Cal.com and Vapi webhook reports.